

Derivation of the q_{β_k} Variational Update

Supervised Bayesian Matrix Factorization

Self-Contained Reference for `code/update_beta.R`

Andrew Walther

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Abstract

This document derives the coordinate-ascent variational inference (CAVI) update for the survival coefficient β_k in the Supervised Bayesian Matrix Factorization (Supervised MF) model. Because β appears only in the Cox proportional-hazards survival likelihood (not in the genomics term), the update depends solely on that likelihood. A second-order Taylor approximation makes the update tractable, and the resulting quadratic objective reduces to a one-dimensional Empirical Bayes Normal Means (EBNM) problem. The EBNM is solved using the `ebnm` R package with a `point_normal` prior, which automatically shrinks irrelevant factors toward zero.

The derivation follows the notation of the parent document `MF_Derivations_UpdateAlgo_REVISED.tex` and maps directly to the standalone R implementation in `code/update_beta.R`. This is the first of four modular derivation documents ($q\beta \rightarrow qL \rightarrow qF \rightarrow q\tau$).

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1 Setup and Notation

1.1 Model

The Supervised MF model combines a genomics matrix factorization with a Cox proportional-hazards survival model:

$$\text{Genomics: } Y_{ij} = \sum_{k=1}^K l_{ik} f_{jk} + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \tau_j^{-1}) \quad (1)$$

$$\text{Survival: } h(t_i) = h_0(t_i) \exp(\eta_i), \quad \eta_i = \sum_{k=1}^K l_{ik} \beta_k = \mathbf{l}_i^\top \boldsymbol{\beta} \quad (2)$$

where Y is $n \times p$, L is $n \times K$ (patient loadings), F is $p \times K$ (biological factors), $\boldsymbol{\beta} \in \mathbb{R}^K$ are survival coefficients, τ_j is the feature-specific noise precision, and (t_i, δ_i) are the observed survival time and event indicator for patient i .

1.2 Key Notation

Throughout this document, $q(\cdot)$ denotes the variational posterior and $g(\cdot)$ denotes the (learned) empirical Bayes prior. The critical distinction between posterior means and second moments:

Symbol	Definition	R variable
$\bar{l}_{ik} = \bar{l}_{ik}$	$\mathbb{E}_q[l_{ik}]$ (posterior mean)	<code>EL[i,k]</code>
$\bar{l}_{ik}^2 = \overline{l_{ik}^2}$	$\mathbb{E}_q[l_{ik}^2]$ (posterior 2nd moment)	<code>EL2[i,k]</code>
$\bar{\beta}_k = \bar{\beta}_k$	$\mathbb{E}_q[\beta_k]$ (posterior mean)	<code>EBeta[k]</code>
$\bar{\beta}_k^2 = \overline{\beta_k^2}$	$\mathbb{E}_q[\beta_k^2]$ (posterior 2nd moment)	<code>EBeta2[k]</code>
$z_i^{-k} = z_i^{-k}$	Partial working response (Section 4)	<code>z_no_k</code>
W_{ii}	Cox neg-diagonal Hessian (positive)	<code>w[i]</code>
z_i	Cox working response $\hat{\eta}_i + u_i/W_{ii}$	<code>z[i]</code>

Verification Check

Second moment identity. For any random variable X with finite variance:

$$\mathbb{E}_q[X^2] = \text{Var}_q(X) + (\mathbb{E}_q[X])^2 \geq (\mathbb{E}_q[X])^2$$

with equality if and only if the posterior is degenerate (zero variance). Whenever we write $\bar{l}_{ik}^2$, this refers to the *full* second moment, not the squared mean. In code: `EL2[i,k] = sd^2 + mean^2` from the EBNM result.

1.3 What is Fixed During the β Update

When updating $q(\beta_k)$, the following quantities are held fixed:

- Posterior moments of loadings: $\bar{l}_{ik}, \bar{l}_{ik}^2$ (from the L update of this iteration)
- Cox Taylor working quantities: z_i, W_{ii} (computed at the start of the outer loop)

- All $\bar{\beta}_{k'}$ and $\bar{\beta}_{k'}^2$ for $k' \neq k$ (fixed within the k-loop)
- Factor moments $\bar{f}_{jk}$, $\bar{f}_{jk}^2$, and precision τ_j (from their own updates)

2 The Objective Function for q_{β_k}

The full Evidence Lower Bound (ELBO) to maximise is:

$$\mathcal{F} = \mathbb{E}_q[\log P(Y, t, \delta \mid L, F, \beta, \tau)] + \sum_k \mathbb{E}_{q_{l_k}} \left[\log \frac{g_{l_k}}{q_{l_k}} \right] + \sum_k \mathbb{E}_{q_{f_k}} \left[\log \frac{g_{f_k}}{q_{f_k}} \right] + \sum_k \mathbb{E}_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}}{q_{\beta_k}} \right] \quad (3)$$

The full data likelihood splits into genomics and survival terms:

$$\log P(Y, t, \delta \mid L, F, \beta, \tau) = \underbrace{\log P(Y \mid L, F, \tau)}_{\text{genomics (no } \beta)} + \underbrace{\log P(t, \delta \mid L, \beta)}_{\text{survival}} \quad (4)$$

Key observation: β does not appear in the genomics term. Therefore, the genomics log-likelihood is *constant* with respect to β_k and drops out of the objective. The objective for q_{β_k} is:

$$\boxed{\mathcal{F}(q_{\beta_k}, g_{\beta_k}) = \mathbb{E}_{q_{l_k}, q_{\beta}} [\log P(t, \delta \mid L, \beta)] + \sum_k \mathbb{E}_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right]} \quad (5)$$

3 Taylor Approximation of the Cox Likelihood

The Cox partial log-likelihood $\ell(\eta)$ is non-conjugate in L and β . We apply a second-order Taylor expansion around the current expansion point $\hat{\eta}_i = \sum_k \bar{l}_{ik} \bar{\beta}_k$:

$$\ell(\eta) \approx \ell(\hat{\eta}) + (\eta - \hat{\eta})^\top \nabla \ell(\hat{\eta}) + \frac{1}{2} (\eta - \hat{\eta})^\top \nabla^2 \ell(\hat{\eta}) (\eta - \hat{\eta}) \quad (6)$$

where the **score** u_i and **neg-diagonal Hessian** W_{ii} are:

$$u_i = \delta_i - \exp(\hat{\eta}_i) \sum_{j: t_j \leq t_i} \frac{\delta_j}{\sum_{m: t_m \geq t_j} \exp(\hat{\eta}_m)} \quad (\text{score}) \quad (7)$$

$$W_{ii} = \exp(\hat{\eta}_i) \sum_{j: t_j \leq t_i} \frac{\delta_j}{\sum_{m: t_m \geq t_j} \exp(\hat{\eta}_m)} \quad (\text{neg-diag Hessian, } W_{ii} > 0) \quad (8)$$

Completing the square in η_i yields the quadratic surrogate:

$$\log P(t, \delta \mid L, \beta) \approx \sum_i \left[-\frac{1}{2} W_{ii} (\eta_i - z_i)^2 \right] + C \quad (9)$$

where the **working response** is:

$$z_i = \hat{\eta}_i + \frac{u_i}{W_{ii}} \quad (10)$$

The full derivation of this approximation is in the parent document (`REVISED.tex`, Section 3). In code: `calc_cox_taylor()`, lines 70–93 of `Supervised_Bayesian_MF_V2.R`.

4 Partial Working Response z_i^{-k}

To update β_k via coordinate ascent, we isolate the dependence on β_k by defining the **partial working response**:

$$z_i^{-k} = z_i^{-k} := z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \quad (11)$$

This removes all contributions from other factors, leaving the residual signal that factor k should explain.

4.1 Reuse Across L and β Updates

Verification Check

z_no_k reuse rationale. Note that z_i^{-k} depends only on $k' \neq k$:

$$z_i^{-k} = z_i - \underbrace{\sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'}}_{\text{no } l_{ik} \text{ or } \beta_k}$$

Therefore, z_i^{-k} is *unchanged* when the L update of factor k modifies $\bar{l}_{ik}$ (since only k -indexed quantities change). This means the *same* z_i^{-k} computed before the L update of factor k can be reused for the β_k update of factor k , saving computation. In the Gauss-Seidel CAVI loop: **z_no_k** is computed once per factor k and reused in both step (a) [L update] and step (c) [β update].

The decomposition $z_i \approx z_i^{-k} + \bar{l}_{ik} \beta_k$ follows immediately from the definition of the working response. In code:

```
eta_no_k <- as.vector(EL %*% EBeta) - EL[, k] * EBeta[k]
z_no_k    <- z - eta_no_k    # = z_i - sum_{k'!=k} l_bar * beta_bar
```

5 Expanding the Survival Likelihood Over $q(L)$

Substituting the working-response form (9) into the objective (5):

$$\mathcal{F}(q_\beta, g_\beta) \approx \mathbb{E}_{q_L, q_\beta} \left[\sum_i -\frac{1}{2} W_{ii} \left(z_i - \sum_k l_{ik} \beta_k \right)^2 \right] + \sum_k \mathbb{E}_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (12)$$

5.1 Expanding the Squared Term

Derivation Steps

Step 4.1: Expand the square.

$$\left(z_i - \sum_k l_{ik} \beta_k \right)^2 = z_i^2 - 2z_i \sum_k l_{ik} \beta_k + \left(\sum_k l_{ik} \beta_k \right)^2 \quad (13)$$

Step 4.2: Expand the cross-product term.

$$\left(\sum_k l_{ik} \beta_k \right)^2 = \sum_k l_{ik}^2 \beta_k^2 + 2 \sum_{k < k'} l_{ik} \beta_k l_{ik'} \beta_{k'} \quad (14)$$

Correction Note

Correction R5 (from REVISED.tex). The document MF_Derivations_UpdateAlgo_2-12-26.pdf (Eq. 68) and also the March 6, 2026 draft (Eq. 68 on p. 7) contain a typographical error where the exponent on β_k is missing: $\sum_k l_{ik}^2 \beta_k$ should be $\sum_k l_{ik}^2 \beta_k^2$. The *final* formulas for A_k and B_k in those documents are correct; only this intermediate step has the typo.

Step 4.3: Apply $\mathbb{E}_{q_l}[\cdot]$ using mean-field independence.

Under mean-field variational inference, $q(L) = \prod_k q_{l_k}(\mathbf{l}_k)$ and $l_{ik} \perp l_{ik'}$ for $k \neq k'$ under q . Therefore:

$$\begin{aligned} \mathbb{E}_{q_l}[l_{ik}] &= \bar{l}_{ik} \\ \mathbb{E}_{q_l}[l_{ik}^2] &= \overline{l_{ik}^2} = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2 \\ \mathbb{E}_{q_l}[l_{ik} l_{ik'}] &= \bar{l}_{ik} \bar{l}_{ik'} \quad (k \neq k') \end{aligned}$$

The cross terms $\mathbb{E}[l_{ik} \beta_k l_{ik'} \beta_{k'}] = \bar{l}_{ik} \bar{l}_{ik'} \beta_k \beta_{k'}$ for $k \neq k'$. These cross-factor terms survive the L expectation but are subsequently absorbed into the z_i^{-k} decomposition.

Step 4.4: Variance-mean decomposition.

After taking $\mathbb{E}_{q_l}$ and applying the decomposition $\overline{l_{ik}^2} = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2$, the squared term splits as:

$$\mathbb{E}_{q_l} \left[\left(\sum_k l_{ik} \beta_k \right)^2 \right] = \underbrace{\left(\sum_k \bar{l}_{ik} \beta_k \right)^2}_{\text{plug-in square}} + \underbrace{\sum_k \text{Var}_q(l_{ik}) \beta_k^2}_{\text{variance correction}} \quad (15)$$

Step 4.5: Reconstitute the full objective.

The expectation of the squared-residual term becomes:

$$\mathbb{E}_{q_l} \left[\left(z_i - \sum_k l_{ik} \beta_k \right)^2 \right] = \left(z_i - \sum_k \bar{l}_{ik} \beta_k \right)^2 + \sum_k \text{Var}_q(l_{ik}) \beta_k^2 \quad (16)$$

Substituting into the objective (12):

$$\begin{aligned} \mathcal{F}(q_\beta, g_\beta) &\approx \mathbb{E}_{q_\beta} \left[\sum_i -\frac{1}{2} W_{ii} \left(z_i - \sum_k \bar{l}_{ik} \beta_k \right)^2 \right] \\ &\quad + \mathbb{E}_{q_\beta} \left[\sum_i -\frac{1}{2} W_{ii} \sum_k \text{Var}_q(l_{ik}) \beta_k^2 \right] \\ &\quad + \sum_k \mathbb{E}_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \end{aligned} \quad (17)$$

6 Coordinate-Ascent for a Single β_k

Apply the mean-field independence $q(\boldsymbol{\beta}) = \prod_k q_{\beta_k}(\beta_k)$ and fix all $\beta_{k'}$ for $k' \neq k$. Substitute $z_i = z_i^{-k} + \bar{l}_{ik}\beta_k$ into the first term of (17):

$$\sum_i -\frac{1}{2}W_{ii} \left(z_i - \sum_k \bar{l}_{ik}\beta_k \right)^2 = \sum_i -\frac{1}{2}W_{ii} \left(z_i^{-k} - \bar{l}_{ik}\beta_k \right)^2 + C \quad (18)$$

where C collects all terms constant in β_k . Expanding the square:

$$-\frac{1}{2}W_{ii} \left(z_i^{-k} - \bar{l}_{ik}\beta_k \right)^2 = -\frac{1}{2}W_{ii} \left[\left(z_i^{-k} \right)^2 - 2z_i^{-k}\bar{l}_{ik}\beta_k + \bar{l}_{ik}^2\beta_k^2 \right] \quad (19)$$

Summing over i , summing over the variance-correction term, and collecting the β_k^2 and β_k coefficients:

Derivation Steps

Identifying the quadratic form.

The objective for a single β_k (up to constants in β_k) is:

$$\mathcal{F}(q_{\beta_k}, g_{\beta_k}) = \mathbb{E}_{q_{\beta_k}} \left[\underbrace{\sum_i W_{ii} z_i^{-k} \bar{l}_{ik}}_{B_k} \beta_k - \frac{1}{2} \underbrace{\sum_i W_{ii} \bar{l}_{ik}^2}_{A_k} \beta_k^2 \right] + \mathbb{E}_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] + C \quad (20)$$

where we have combined the plug-in squared-mean term $\bar{l}_{ik}^2$ with the variance correction $\text{Var}_q(l_{ik})$ into the full second moment $\overline{l_{ik}^2} = \bar{l}_{ik}^2 + \text{Var}_q(l_{ik})$:

$$\bar{l}_{ik}^2 + \text{Var}_q(l_{ik}) = \overline{l_{ik}^2} \quad (21)$$

This yields the precision and signal coefficients:

$$A_k = \sum_{i=1}^n W_{ii} \overline{l_{ik}^2} \quad (22)$$

$$B_k = \sum_{i=1}^n W_{ii} z_i^{-k} \bar{l}_{ik} \quad (23)$$

Verification Check

Sign and dimension checks.

- $A_k > 0$: since $W_{ii} > 0$ (floored at 10^{-6}) and $\overline{l_{ik}^2} \geq 0$, we have $A_k \geq 0$. A precision floor $A_k \geq 10^{-10}$ is applied in code to prevent division by zero.
- A_k is a **scalar** (sum over $i = 1, \dots, n$).
- B_k is a **scalar** (sum over $i = 1, \dots, n$).
- The quadratic $-\frac{1}{2}A_k\beta_k^2 + B_k\beta_k$ is concave in β_k (since $A_k > 0$), as required for a valid EBNM likelihood.

7 The EBNM Problem

The objective (20) is exactly a one-dimensional EBNM problem: maximise over $(q_{\beta_k}, g_{\beta_k})$ the objective $\mathbb{E}_q[-\frac{1}{2A_k}(\beta_k - x_k)^2] + \mathbb{E}_q[\log(g/q)]$ where:

EBNM Problem

EBNM Inputs:

$$x_k = \frac{B_k}{A_k} = \frac{\sum_i W_{ii} z_i^{-k} \bar{l}_{ik}}{\sum_i W_{ii} \bar{l}_{ik}^2} \quad (24)$$

$$s_k = \frac{1}{\sqrt{A_k}} = \frac{1}{\sqrt{\sum_i W_{ii} \bar{l}_{ik}^2}} \quad (25)$$

EBNM Call:

$$(\hat{g}_{\beta_k}, \hat{q}_{\beta_k}) = \text{EBNM}(x_k, s_k)$$

Posterior Extraction:

$$\bar{\beta}_k = \hat{q}_{\beta_k}.\text{posterior}\$mean \quad (26)$$

$$\overline{\beta_k^2} = \hat{q}_{\beta_k}.\text{posterior}\$sd^2 + \hat{q}_{\beta_k}.\text{posterior}\$mean^2 \quad (27)$$

7.1 Prior Variance Note

The derivation with a fixed Gaussian prior $g(\beta_k) \sim \mathcal{N}(0, \sigma^2)$ would yield an additional term $1/\sigma^2$ in A_k :

$$A_k^{\text{fixed prior}} = \sum_i W_{ii} \bar{l}_{ik}^2 + \frac{1}{\sigma^2} \quad (28)$$

In the implementation, σ^2 is *not* fixed in advance. Instead, `prior_family = "point_normal"` causes the EBNM solver to *empirically estimate* the prior from data (empirical Bayes). This is equivalent to marginalising over σ^2 , and the $1/\sigma^2$ term is absorbed into the EBNM prior estimation. See REVISED.tex lines 911–914.

7.2 Error-in-Variables Interpretation

Verification Check

Error-in-variables correction. The use of $\bar{l}_{ik}^2$ (full second moment) rather than $\bar{l}_{ik}^2$ (squared mean) in A_k encodes an *error-in-variables correction*:

1. Higher posterior variance $\text{Var}_q(l_{ik})$ increases $\bar{l}_{ik}^2 = \bar{l}_{ik}^2 + \text{Var}_q(l_{ik})$.
2. This increases A_k , which decreases both $x_k = B_k/A_k$ and $s_k = 1/\sqrt{A_k}$. Crucially, the ratio $|x_k|/s_k = |B_k|/\sqrt{A_k}$ decreases.
3. A smaller $|x_k|/s_k$ ratio means the signal is weaker relative to the noise, so the EBNM

prior applies more shrinkage.

4. Net effect: more uncertainty in $L \Rightarrow$ more shrinkage of β_k toward zero. This prevents overfitting β to uncertain loadings.

Mathematically: if $\text{Var}_q(l_{ik}) = 0$ for all i , then $A_k = \sum_i W_{ii} \bar{l}_{ik}^2$ and the EBNM with a flat prior recovers the weighted least squares estimator:

$$\hat{\beta}_k^{\text{WLS}} = \frac{\sum_i W_{ii} z_i^{-k} \bar{l}_{ik}}{\sum_i W_{ii} \bar{l}_{ik}^2}$$

This is verified in test T2.1 of `tests/test_update_beta.R`.

8 Verification and Self-Consistency Checks

Verification Check

Check 1: WLS limit. When posterior variance is zero ($\bar{l}_{ik}^2 = \bar{l}_{ik}^2$) and the EBNM prior is flat (non-informative), the posterior mean equals the WLS estimator $x_k = B_k/A_k$. **Confirmed in test T2.1.**

Verification Check

Check 2: Null factor shrinkage. When z_i^{-k} is pure noise (no signal from factor k), $B_k \approx 0$ and $x_k \approx 0$. The point-normal prior shrinks $\hat{\beta}_k$ to approximately zero. **Confirmed in tests T5.1–T5.2.**

Verification Check

Check 3: Error-in-variables direction. Larger $\bar{l}_{ik}^2$ (more posterior variance in L) yields larger A_k and a smaller ratio $|B_k/A_k|/s_k = |B_k|/\sqrt{A_k}$, leading to more shrinkage. Note: $s_k = 1/\sqrt{A_k}$ is *smaller* (not larger) when A_k is larger. The shrinkage effect comes from the reduction of the pseudo-observation x_k , not from an increase in s_k . **Confirmed in test T6.1.**

Verification Check

Check 4: Consistency with V2.R. The modular function `update_beta_k()` produces identical results (to machine precision) as the inline code in `Supervised_Bayesian_MF_V2.R` lines 356–361 for all tested inputs. **Confirmed in test T9.1.**

Verification Check

Check 5: z_no_k reuse invariance. z_i^{-k} does not depend on l_{ik} or β_k . Therefore the value computed before the L update of factor k remains valid for use in the β_k update. **Confirmed analytically and in test T8.1.**

9 Code Mapping

Code Mapping (update_beta.R)

Math $\rightarrow$ Code correspondence for update_beta.R:

Math	R Variable	Computation
W_{ii}	w	calc_cox_taylor(eta,time,status)\$w
z_i	z	eta + u/w
z_i^{-k}	z_no_k	compute_z_no_k(z, EL, EBeta, k)
$\bar{l}_{ik}$	EL[i,k]	res_L\$posterior\$mean
$\bar{l}_{ik}^2$	EL2[i,k]	sd^2 + mean^2
A_k	res\$A	max(sum(w * EL2.k), 1e-10)
B_k	res\$B	sum(w * z_no_k * EL.k)
x_k	res\$x	B.k / A.k
s_k	res\$s	1 / sqrt(A.k)
$\bar{\beta}_k$	res\$mean	ebnm_result\$posterior\$mean
$\bar{\beta}_k^2$	res\$second	sd^2 + mean^2

Pseudocode matching V2.R lines 356–361:

```

A_k  <- max(sum(w * EL2[, k]), 1e-10)      # precision
B_k  <- sum(w * z_no_k * EL[, k])          # signal
x_k  <- B_k / A_k                          # pseudo-obs
s_k  <- 1 / sqrt(A_k)                      # pseudo-noise
res  <- ebnm(x = x_k, s = s_k,
             prior_family = "point_normal")
EBeta[k] <- res$posterior$mean             # posterior mean
EBeta2[k] <- res$posterior$sd^2 + mean^2   # 2nd moment

```

10 Algorithm Summary

The complete q_{β_k} update algorithm (Gauss-Seidel over $k = 1, \dots, K$):

1. **Prerequisite:** Compute Cox Taylor quantities (z_i, W_{ii}) from $\text{calc_cox_taylor}(\hat{\eta}, t, \delta)$.
2. **For** $k = 1, \dots, K$:
 - (a) Compute partial working response: $z_i^{-k} = z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'}$
 - (b) Compute precision: $A_k = \max\left(\sum_i W_{ii} \bar{l}_{ik}^2, 10^{-10}\right)$
 - (c) Compute signal: $B_k = \sum_i W_{ii} z_i^{-k} \bar{l}_{ik}$
 - (d) Form EBNM inputs: $x_k = B_k / A_k$, $s_k = 1 / \sqrt{A_k}$
 - (e) Solve: $(\hat{g}_{\beta_k}, \hat{q}_{\beta_k}) = \text{EBNM}(x_k, s_k)$
 - (f) Extract: $\bar{\beta}_k \leftarrow \hat{q}.\text{posterior}\$mean$; $\bar{\beta}_k^2 \leftarrow \hat{q}.\text{posterior}\$sd^2 + \text{mean}^2$
 - (g) **Gauss-Seidel:** use updated $\bar{\beta}_k$ when computing $z_i^{-k'}$ for subsequent $k' > k$

A Notation Index

Symbol	Dimension	Meaning
n	scalar	number of patients
p	scalar	number of genomic features
K	scalar	number of latent factors
Y	$n \times p$	observed genomics data
L	$n \times K$	patient loading matrix
F	$p \times K$	biological factor matrix
β	$K \times 1$	survival coefficient vector
τ_j	p -vector	feature-specific noise precision
(t_i, δ_i)	n -vectors	survival time and event indicator
η_i	n -vector	linear predictor $\mathbf{l}_i^\top \beta$
u_i	n -vector	Cox score
W_{ii}	n -vector	Cox neg-diagonal Hessian (positive)
z_i	n -vector	working response $\hat{\eta}_i + u_i/W_{ii}$
z_i^{-k}	n -vector	partial working response (excludes factor k)
A_k	scalar	EBNM precision for factor k
B_k	scalar	EBNM signal for factor k
x_k	scalar	EBNM pseudo-observation B_k/A_k
s_k	scalar	EBNM pseudo-noise $1/\sqrt{A_k}$

B Cross-References to REVISED.tex

This document	REVISED.tex
Eq. (1)–(2)	Sec. 1A (model specification)
Eq. (3)	Eq. (1) (full ELBO)
Eq. (7)–(10)	Sec. 3 (Taylor expansion)
Eq. (22)–(23)	Eqs. (79)–(80) (A_k , B_k)
Eq. (24)–(25)	Eqs. (81)–(82) (EBNM inputs)
Correction R5	Sec. 1 (revision log)